

Programme	B. Sc. Mathematics Honours			
Course Code	MAT6CJ305 / MAT8MN305			
Course Title	ELEMENTARY NUMBER THEORY			
Type of Course	Major			
Semester	VI			
Academic Level	300-399			
Course Details	Credit	Lecture/Tutorial per week	Practicum per week	Total Hours
	4	4	-	60
Pre-requisites	Arithmetic of integers, basic set theory and proof techniques.			
Course Summary	We start number theory with the division algorithm, g.c.d., and the Euclidean algorithm for computing it, essential for solving Diophantine equations like $ax + by = c$. We then prove the Fundamental Theorem of Arithmetic, discuss the infinitude of primes and the sieve of Eratosthenes. Following that, we cover Linear Congruences, the Chinese Remainder theorem, and Fermat's Little Theorem. Finally, we explore Wilson's Theorem, Euler's Phi Function, and Euler's Theorem.			

Course Outcomes:

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Apply the division algorithm and Euclidean algorithm to compute greatest common divisors (gcd) and solve related divisibility problems.	Ap	C	Internal Exam/ Assignment/ Seminar/ Viva/ End Sem Exam
CO2	Solve Diophantine equations for integer solutions, deduce prime factorization through the fundamental theorem of arithmetic, and identify prime numbers using the sieve of Eratosthenes.	Ap	C	Internal Exam/ Assignment/ Seminar/Viva/ End Sem Exam
CO3	Apply the properties of congruence and the Chinese Remainder Theorem to solve systems of linear congruences.	Ap	C	Internal Exam/ Assignment/ Seminar/ Viva/ End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Textbook		Elementary Number Theory, David Burton, M, Seventh Edition, McGraw – Hill (2007).		
Module	Unit	Content	Hrs (60)	External Marks (70)
I	Module I		12	Min.15
	1	Section 2.2 The division algorithm (proof of theorem 2.1 omitted).		
	2	Section 2.3 The greatest common divisor - up to and including theorem 2.3 and its corollary.		
	3	Section 2.3 The greatest common divisor - All topics from definition 2.3 onwards.		
	4	Section 2.4 The Euclidean algorithm - up to Theorem 2.7.		
	5	Section 2.4 The Euclidean algorithm - All topics from Theorem 2.7 onwards.		
II	Module II		11	Min.15
	6	Section 2.5 The Diophantine equation $ax+by = c$ - up to and including Theorem 2.9.		
	7	Section 2.5 - All topics from Example 2.4 onwards.		
	8	Section 3.1 The fundamental theorem of arithmetic - up to Theorem 3.2.		
	9	Section 3.1 The fundamental theorem of arithmetic - All topics from Theorem 3.2 onwards.		
	10	Section 3.2 The sieve of Eratosthenes (up to and including theorem 3.4 only)		
III	Module III			

	11	Section 4.2 Basic properties of congruence - up to Theorem 4.2.	13	Min.15
	12	Section 4.2 Basic properties of congruence - All topics from Theorem 4.2 onwards.		
	13	Section 4.4 Linear congruences and the Chinese remainder theorem - up to Theorem 4.8.		
	14	Section 4.4 Linear congruences and the Chinese remainder theorem - All Topics from Theorem 4.8 (proof of Theorem 4.8 omitted).		
	15	Section 5.2 Fermat's little theorem and pseudo primes - up to Lemma. (omit a different proof for Fermat's theorem)		
	16	Section 5.2 Fermat's little theorem and pseudo primes - All topics from Lemma onwards.		
IV	Module IV		12	Min.15
	17	Section 5.3 Wilson's theorem - Up to Theorem 5.5.		
	18	Section 5.3 Wilson's theorem - All topics from Theorem 5.5 onwards.		
	19	Section 7.2 Euler's phi-function - up to Lemma.		
	20	Section 7.2 Euler's phi-function - All Topics from Lemma onwards. (proof of Theorem 7.2 omitted).		
	21	Section 7.3 Euler's theorem. (Second proof of Euler's theorem omitted).		
	22	Section 7.4 Some properties of the phi-function (Proof of Theorem 7.8 omitted).		
V	Module V (Open Ended)			

	Proof of Theorem 4.8. Chinese Remainder Theorem and remaining portions of Section 4.4 Section 6.1 The sum and the number of divisors Linear congruences and the Chinese remainder theorem. Section 6.3 The Greatest Integer Function - up to Theorem 6.11.	12	
References 1. Rosen, Kenneth H. <i>Elementary number theory</i>. London: Pearson Education, 2011. 2. Eynden, Charles Vanden. <i>Elementary number theory</i>. Waveland Press, 2006. 3. Gehring, F. W., and P. R. Halmos. <i>Graduate Texts in Mathematics</i>, 1976. 4. Hsiung, C. Y. <i>Elementary theory of numbers</i>. World Scientific, 1992. 5. Hoffman P., <i>The man who loved only numbers: The story of Paul Erdős and the search for mathematical truth</i>, Little Brown & Company, 1999.			

***70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.**

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	1	1	0	0	3	0	3	0	3	0	0
CO 2	1	1	0	0	3	0	3	0	3	0	0
CO 3	0	0	1	0	3	0	3	0	3	0	0

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓